4. The molar conductivity of a 0.5 mol dm^{-3} solution of $AgNO_3$ with electrolytic conductivity of 5.76×10^{-3} S cm^{-1} at 298 K is

a) 2.88 S $cm^2 mol^{-1}$ b) 11.52 S $cm^2 mol^{-1}$

c) 0.086 S $cm^2 mol^{-1}$ d) 28.8 S $cm^2 mol^{-1}$

5.

Electrolyte	KCl	KNO_3	HCl	NaOAC	NaCl
Λ_m (S cm^2 mol^{-1})	149.9	145.0	426.2	91.0	126.5

Calculate Λ°_{HOAC} using appropriate molar conductances of the electrolytes listed above at infinite dilution in water at $25^{\circ}C$.

a) 517.2 b) 552.7 c) 390.7 d) 217.5

6. Faraday constant is defined as

a) charge carried by 1 electron

b) charge carried by one mole of electrons

c) charge required to deposit one mole of substance

d) charge carried by 6.22×10^{10} electrons.

7. How many faradays of electricity are required for the following reaction to occur $MnO_4^- \rightarrow Mn^{2+}$

a) 5F b) 3F c) 1F d) 7F

8. A current strength of 3.86 A was passed through molten Calcium oxide for 41minutes and 40 seconds. The mass of Calcium in grams deposited at the cathode is (atomic mass of Ca is 40g / mol and 1F = 96500C).

a) 4 b) 2 c) 8 d) 6

9. During electrolysis of molten sodium chloride, the time required to produce 0.1mole of chlorine gas using a current of 3A is

a) 55 minutes b) 107.2 minutes c) 220 minutes d) 330 minutes

10. The number of electrons delivered at the cathode during electrolysis by a current of 1A in 60 seconds is (charge of electron = 1.6×10^{-19} C)

a) 6.22×10^{23} b) 6.022×10^{20} c) 3.75×10^{20} d) 7.48×10^{23}

11. Which of the following electrolytic solution has the least specific conductance

a) 2N b) 0.002N c) 0.02N d) 0.2N

12. While charging lead storage battery

a) $PbSO_4$ on cathode is reduced to Pb b) $PbSO_4$ on anode is oxidised to PbO_2

c) $PbSO_4$ on anode is reduced to Pb d) $PbSO_4$ on cathode is oxidised to Pb

XII U9 Electro Chemistry.indd 63 11-01-2021 14:44:40